A Differential Equation Approach to Continuous-Time Simple Symmetric Random Walks

Mathematics IV Presentation

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Model Setup: Continuous-Time Simple Symmetric Random Walk

- We consider a stochastic process $\{X(t)\}_{t\geq 0}$ on $\mathbb{Z}$ with initial state $X(0) = 0$.
- **Waiting Time:** Time between jumps

$$\tau \sim \text{Exp}(2\lambda)$$

- **Jump Rule:** At each jump

$$\xi \in \{-1, +1\}, \quad \mathbb{P}(\xi = \pm 1) = \frac{1}{2}$$

- **State Update:**

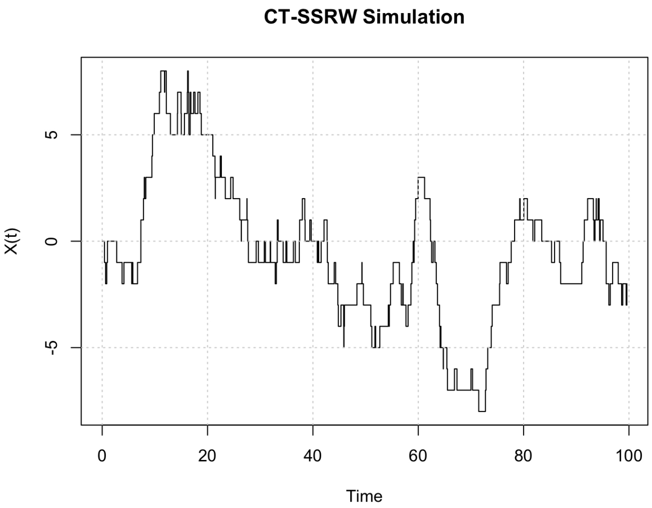
$$X(t + \tau) = X(t) + \xi$$

- **Number of Jumps:**

$$N(t) \sim \text{Poisson}(2\lambda t)$$

- **Interpretation:** Jumps occur randomly in time, and each jump moves the process one step left or right.

Simulation of the Process



- The process evolves through random jumps in continuous time
- Each jump is ± 1 with equal probability.
- **This process is known as a Markov Jump Process.**
- **Here, $\lambda = 1$.**

Kolmogorov Forward Equation

- Let $p_i(t) = \mathbb{P}(X(t) = i)$ denote the probability of being in state i at time t.
- The evolution of probabilities is governed by:

$$\frac{dp_i(t)}{dt} = \lambda p_{i-1}(t) + \lambda p_{i+1}(t) - 2\lambda p_i(t)$$

- **Interpretation:**
  - Inflow from neighboring states $i - 1$ and $i + 1$
  - Outflow from state i at total rate 2λ
- **Goal:** Solve this system of differential equations to obtain

$$p_i(t) \quad \text{as a function of time}$$

Generating Function Approach

- Define the generating function:

$$G(z,t) = \sum_{i=-\infty}^{\infty} p_i(t)\, z^i$$

- Differentiate with respect to time:

$$\frac{\partial G}{\partial t} = \sum_i \frac{dp_i(t)}{dt} z^i$$

- Substitute the Kolmogorov forward equation:

$$\frac{\partial G}{\partial t} = \sum_i \left[\lambda p_{i-1}(t) + \lambda p_{i+1}(t) - 2\lambda p_i(t)\right] z^i = \lambda(z + z^{-1} - 2)\, G$$

- Rearranging:

$$\frac{1}{G}\frac{\partial G}{\partial t} = \lambda(z + z^{-1} - 2)$$

Solving the Generating Function

- From previous slide:

$$\frac{1}{G}\frac{\partial G}{\partial t} = \lambda(z + z^{-1} - 2)$$

- Using separation of variables:

$$\int \frac{1}{G}\, dG = \int \lambda(z + z^{-1} - 2)\, dt$$

- Integrating:

$$\ln G = \lambda(z + z^{-1} - 2)t + C(z)$$

- Using initial condition $G(z, 0) = 1$:

$$C(z) = 0$$

- Final solution:

$$G(z, t) = e^{\lambda(z + z^{-1} - 2)t}$$

Bessel Generating Function and Solution

- Recall:

$$G(z,t) = e^{\lambda(z+z^{-1}-2)t} = e^{-2\lambda t}\, e^{\lambda t(z+z^{-1})}$$

- **Bessel generating function:**

$$e^{x(z+z^{-1})} = \sum_{i=-\infty}^{\infty} I_i(2x)\, z^i$$

- Using $x = \lambda t$:

$$G(z,t) = e^{-2\lambda t} \sum_{i=-\infty}^{\infty} I_i(2\lambda t)\, z^i$$

- Comparing coefficients of z^i:

$$p_i(t) = e^{-2\lambda t} I_i(2\lambda t)$$

Closed Form Solution

$$p_i(t) = e^{-2\lambda t} I_i(2\lambda t)$$

- $I_i(\cdot)$ denotes the modified Bessel function of the first kind.
- This gives the exact probability of being in state i at time t.

Asymptotic Distribution (CLT)

- Representation:

$$X(t) = S_{N(t)}, \quad N(t) \sim \text{Poisson}(2\lambda t)$$

- For large t:

$$N(t) \approx 2\lambda t$$

- By CLT:

$$S_n \approx \mathcal{N}(0, n)$$

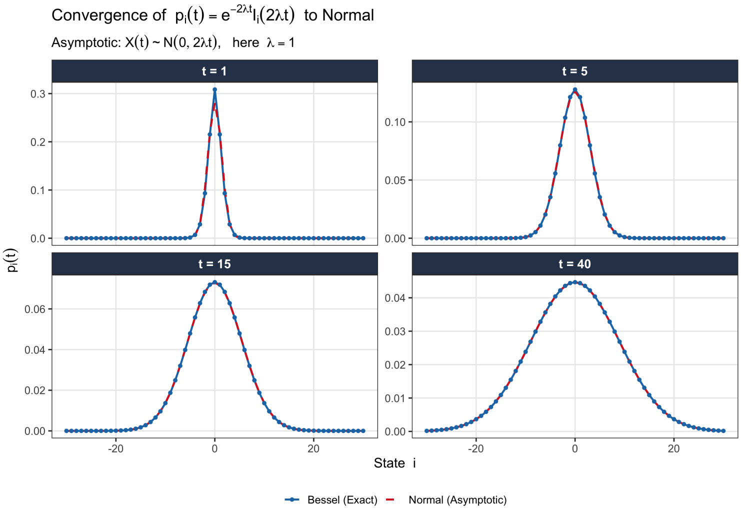
- Hence:

$$X(t) \approx \mathcal{N}(0, 2\lambda t)$$

- Final approximation:

$$p_i(t) \approx \frac{1}{\sqrt{4\pi\lambda t}} e^{-i^2/(4\lambda t)}$$

Convergence to Gaussian Distribution



- Exact distribution:

$$p_i(t) = e^{-2\lambda t} I_i(2\lambda t)$$

- As t increases:
  - Distribution spreads
  - Approaches Gaussian
- Asymptotic limit:

$$X(t) \approx \mathcal{N}(0, 2\lambda t)$$

Application: M/M/1 Queue

- Let $Q(t)$ denote the number of customers in the system at time t.
- **Arrival process:** Poisson with rate λ

$$k \to k+1 \quad \text{at rate } \lambda$$

- **Service process:** Exponential with rate μ

$$k \to k-1 \quad \text{at rate } \mu \;\; (k \geq 1)$$

- **State space:** $\{0, 1, 2, \dots\}$ with boundary at 0
- **Interpretation:** A single-server queue with random arrivals and exponential service times.

Connection to M/M/1 Model

- Recall the Kolmogorov forward equation for CT-SSRW:

$$\frac{dp_i(t)}{dt} = \lambda p_{i-1}(t) + \lambda p_{i+1}(t) - 2\lambda p_i(t)$$

- For the M/M/1 queue:

$$\frac{dp_k(t)}{dt} = \lambda p_{k-1}(t) + \mu p_{k+1}(t) - (\lambda + \mu) p_k(t)$$

- When $\lambda = \mu$:

$$\frac{dp_k(t)}{dt} = \lambda p_{k-1}(t) + \lambda p_{k+1}(t) - 2\lambda p_k(t)$$

- **Observation:** The M/M/1 model behaves like a symmetric random walk in the interior.
- **Key Difference:** The queue is constrained to $k \geq 0$ (reflecting boundary at 0).

Connection to Modern Research

- **Neural Markov Jump Processes**
  Seifner & Sanchez, ICML 2023
- **Problem Setting:**
  - Observe noisy time-series data from a stochastic system
  - Transition rates of the underlying process are unknown
- **Goal:**
  - Learn the transition rates of a Markov Jump Process
  - Infer hidden state dynamics from data
- **Approach:**
  - Parameterize transition rates using neural networks
  - Solve the master equation:
    $$\frac{d}{dt}p(t) = p(t)F$$
  - Optimize to match observed data

How Our Results Support This Framework

- **Our Result:**

$$p_i(t) = e^{-2\lambda t} I_i(2\lambda t)$$

Exact solution of the master equation for a specific generator F

- **Benchmark and Validation**
    - Exact probabilities $p_i(t)$ and known rate λ
    - Generate synthetic data and compare with learned $\hat{p}_i(t)$

We provide an analytical solution for a special case of their problem.

Thank You